REVISION 1.0
4/15/2021

ENGINE CLEANING GUIDE

Review the safety section at the end of this guide before performing engine cleaning.
Use of appropriate PPE is required when performing this job.

Customers love clean engines. This cleaning goes a long way in earing customer trust and satisfaction.

- Gather the following materials. Painting suit, respirator, ear protection, face shield, and gloves, engine washing sprayer, 1 gallon can of mineral spirits, thin painting plastic, blue painters tape, and acquire red and white rags.

- Protect aircraft from chemical over spray by adding plastic covering to windshield and side windows, cover nose tire with trash bag or plastic. Place exhaust stacks covers on, place propeller blade covers on. If not already accomplished cover inlets to air conditioning system. Place absorbent mats on ground for length of the engine. Move air hose to aircraft.

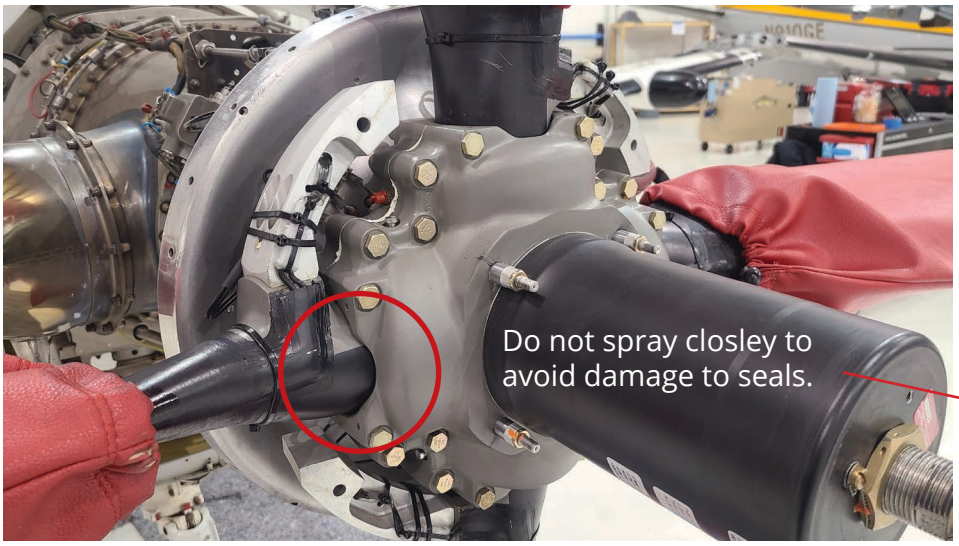
Engine cleaning is accomplished in two steps, first loosen all the excess dirt and grease with spraying mineral spirits, secondly wipe down all surfaces with simple green and white rags. To clean the engine it is most effective to start on the bottom of the engine to remove the worst grease, then moving from one side of the engine to the other. This will help assist you in missing less spots. Then repeat this movement while wiping down the engine.

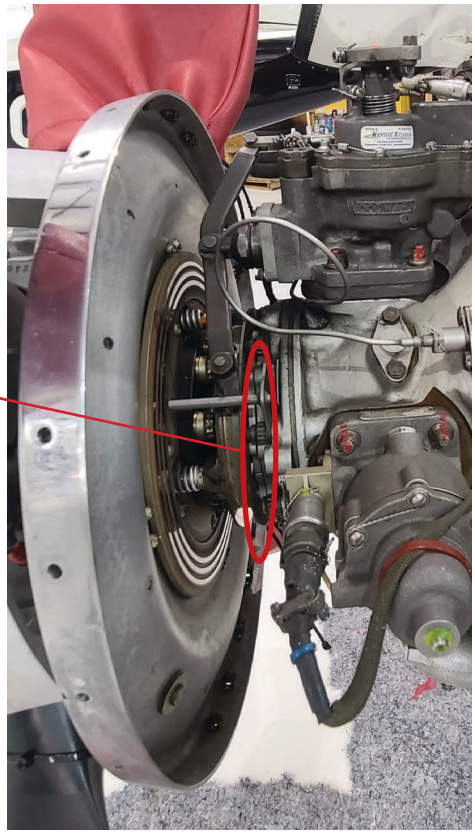
⚠ Verify that the aircraft does not have a battery installed or ground power applied. Remove GPU connection if installed. (Older aircraft when powered have a shunt near the GPU receptacle which will heat up and cause a fire hazard when combined with the atomized mineral spirits). Mineral spirits is used in some applications as starting fluid and is combustible.

1. Begin cleaning with red rags wiping away excess grease from nose gear steering linkages and also nose gear up lock springs.

2. Using the engine cleaning wand and remove remaining grease from nose gear steering and springs and attempt to dissolve any grease clumps. Then spray bottom of engine (Accessory gearbox) nose gear doors and bottom of cowling.

3. Clean the tops of the cowlings, including moving latches back and forth while spraying to clean mechanism.





4. Starting from the LH side clean the top of the engine to the bottom then move forward towards the propeller.

5. Do not spray closely the propeller to engine flange or the propeller blade roots to prevent blowing out the o-rings.

6. Clean propeller hub with mineral spirit spray but from a distance of at least 12 inches.

7. Clean from top to bottom RH side of engine and then pay special attention to the engine control Linkages at the FCU and CAM box.

8. Clean engine fire wall and accessories.

9. Clean engine tubular mount thoroughly including wiring and hoses adjacent.

10. Clean bottom of cowling and nose landing gear.

11. Re-clean nose gear steering linkages to catch anything you missed. This area is usually very dirty.

12. Clean bottom of engine and nose gear doors again

Re-wet the engine by spraying over areas you haven't sprayed since the beginning. Do not air dry the engine with compressed air, it is much easier to wipe down the engine if the mineral spirits is still on the surface.

Wipe off entire engine with white terry towels and simple green. Do not leave wet simple green on surface or it will become sticky over time. Use acid brushes to remove dirt around rivet lines or hard to reach areas.

There are a lot of small parts and surface area, each person should develop their own method of cleaning the engine consistently. Attention to detail comes in handy with this job. The engine wont ever be perfectly clean but it should not appear dirty to a customer.

SAFETY: Information from the Mineral Spirits safety data sheet (SDS):

INGESTION (SWALLOWING): May be harmful if swallowed. May cause throat irritation, nausea, vomiting, Diarrhea and central nervous system effects as noted under
INHALATION (BREATHING). Aspiration hazard: breathing product into the lungs during ingestion or vomiting may cause lung injury and possible death.

MEDICAL CONDITIONS AGGRAVATED BY EXPOSURE: Individuals with pre-existing respiratory tract (nose, throat, and lungs), central nervous system, eye, and/or skin disorders may have increased susceptibility to the effects of exposure.

CHRONIC: Prolonged or repeated inhalation may cause toxic effects as noted under INHALATION (BREATHING). Prolonged or repeated eye contact may cause inflammation of the membrane lining the eyelids and covering the eyeball (conjunctivitis). Prolonged or repeated skin contact may cause drying, cracking, redness, itching, and/or swelling (dermatitis)

Fire information: Paint Thinner (mineral spirits), Paint thinner or mineral spirits is a complex petroleum distillate; flash point range is listed as 104 degrees F. Ignition temperature 473 degrees F, toxicity (3). A category of clear, combustible liquid having petroleum type odor. Mineral spirits are midrange petroleum distillates which is present in many paint thinners, oil base stains, dry cleaning solvents, and some brands of charcoal starter fluids.